

WEEK 3: DATA COLLECTION, DATA CLEANING, AND EXPLORATORY DATA ANALYSIS

Course: Machine Learning and Natural Language Processing

Target Learners: Graduating Computer Science Students

Lecture Duration: 2 Hours | Laboratory Duration: 3 Hours

Course Context: Thesis-Support Course

I. Introduction

Machine Learning (ML) and Natural Language Processing (NLP) projects are fundamentally data-driven. Regardless of how advanced an algorithm is, the quality and appropriateness of its output depend heavily on the quality, relevance, and preparation of the data used for training, testing, analysis, or knowledge retrieval.

A common misconception among beginning AI researchers is that the most important part of an AI project is selecting a sophisticated algorithm. In practice, a substantial portion of AI development involves understanding, collecting, organizing, cleaning, and analyzing data.

For graduating Computer Science students, this topic is particularly important because the datasets used in laboratory activities should, whenever possible, directly support the development of their undergraduate theses.

By the end of this week, students should be able to answer three fundamental questions:

- Do I have the right data for my thesis problem?
- Is my data sufficiently clean and usable for Machine Learning or NLP?
- What does my data reveal before I build an AI model?

II. Specific Learning Outcomes

At the end of the week, the students should be able to:

Knowledge and Understanding

1. Explain the role of data collection, data cleaning, and exploratory data analysis in the Machine Learning and NLP development lifecycle.
2. Differentiate between primary and secondary data and identify appropriate sources of datasets for AI research.
3. Describe common data quality problems, including missing values, duplicate records, inconsistent values, noise, and outliers.
4. Explain the purpose of Exploratory Data Analysis in understanding dataset characteristics and identifying potential problems before model development.

Application and Analysis

1. Identify the data requirements of their proposed thesis project.

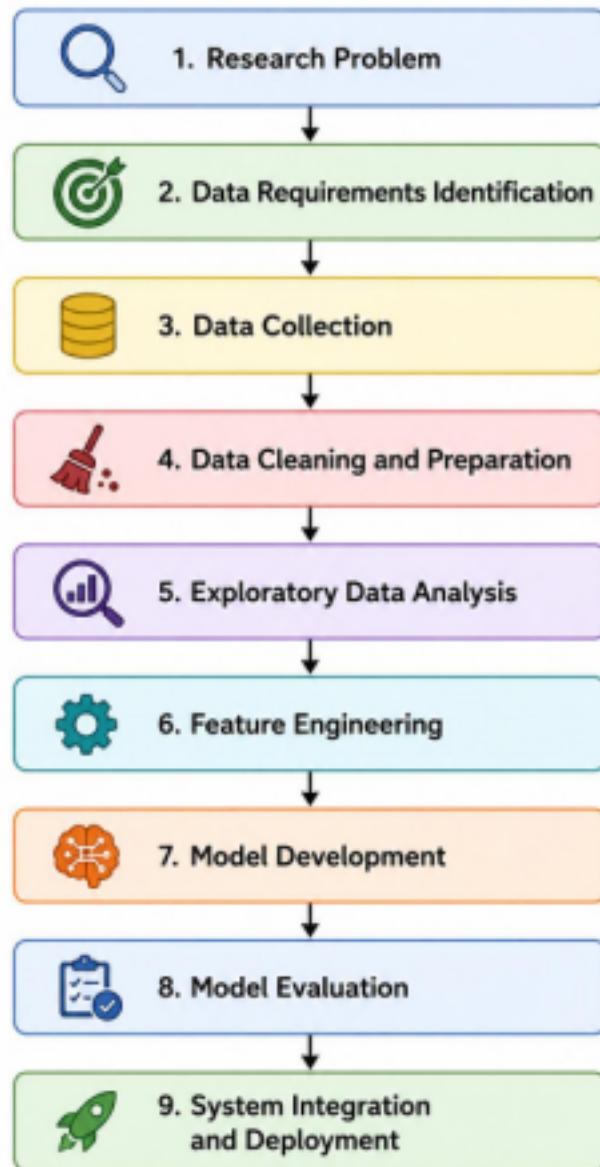
2. Select an appropriate data collection method based on the research problem and proposed AI application.
3. Inspect and clean a dataset using appropriate preprocessing techniques.
4. Perform basic exploratory data analysis using descriptive statistics and visualizations.
5. Identify potential features, target variables, classes, or textual attributes relevant to their proposed ML or NLP task.
6. Analyze potential data quality, privacy, bias, and ethical issues associated with their dataset.

Thesis-Oriented Outcome

Develop a preliminary thesis dataset and data readiness report that can serve as the foundation for subsequent model development activities.

III. Overview of the AI Data Lifecycle

Before developing an AI model, researchers typically follow a data-oriented workflow:



This week's lesson focuses on three critical stages: (1) Data Collection, (2) Data Cleaning, and (3) Exploratory Data Analysis. These stages are interconnected and often iterative rather than strictly linear.

PART I: DATA COLLECTION

IV. Understanding Data Collection

Data collection is the systematic process of gathering information required to answer a research question, develop a computational model, or support a decision-making process. In Machine Learning, data serves as the basis from which a model learns patterns.

Before collecting data, students must first understand: What information does my proposed system need in order to perform its intended task?

V. Identifying Data Requirements

Students should not immediately collect any available dataset. They must first define their data requirements.

1. What is the problem?
2. What should the system do?
3. What information might help make a prediction, classification, analysis, or decision?
4. What should the model predict, classify, generate, or retrieve?

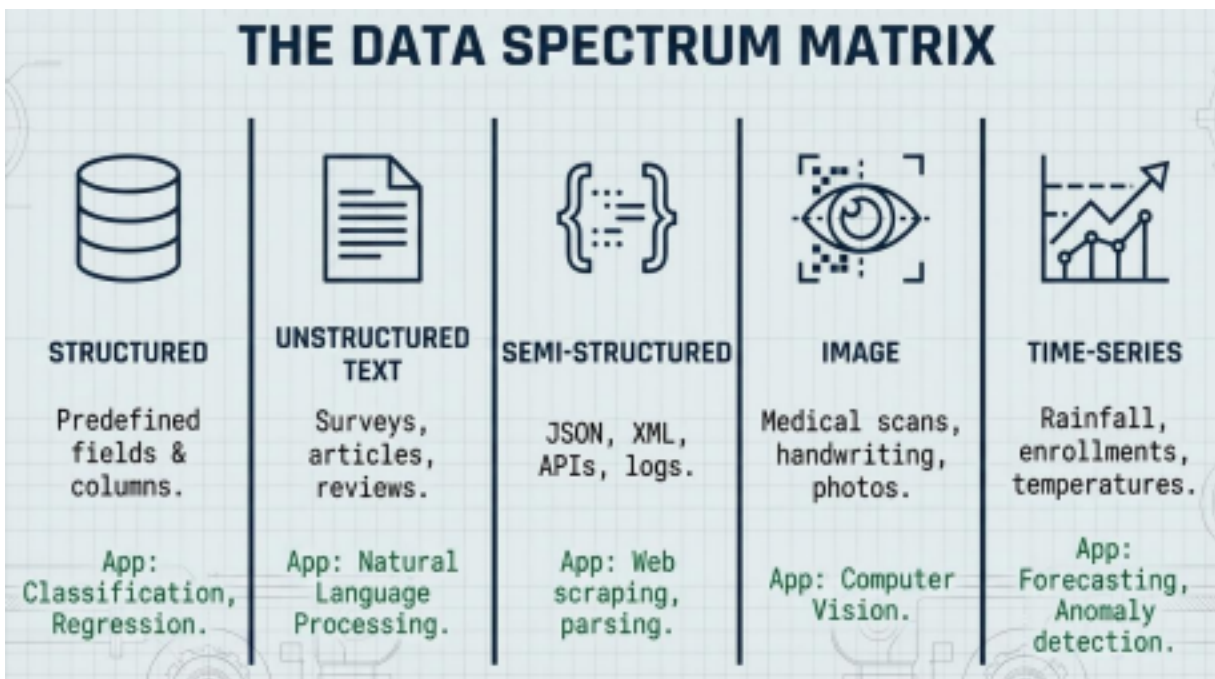
The predicted variable is commonly called the target variable, dependent variable, label, or output variable.

Features and Target Variables

Features are input variables used by a Machine Learning model to identify patterns. The target variable is the value or category the model attempts to predict.

Students should be able to complete the statement: My proposed AI system will use _____ as input in order to predict/classify/generate/retrieve _____.

VI. Types of Data



1. Structured Data

Structured data is organized into predefined fields or columns. Examples include databases, spreadsheets, and CSV files. It is commonly used for classification, regression, prediction, and clustering.

2. Unstructured Text Data

Examples include survey responses, social media posts, student feedback, documents, news articles, and customer reviews. This type of data is commonly used in NLP.

3. Semi-Structured Data

Semi-structured data contains some organizational structure but does not follow a traditional relational database format. Examples include JSON, XML, API responses, and log files.

4. Image Data

Examples include plant leaf images, medical images, handwritten characters, and other visual data used in Computer Vision.

5. Time-Series Data

Time-series data is collected over time. Examples include daily rainfall, monthly enrollment, and temperature records. Common applications include forecasting, trend analysis, and anomaly detection.

VII. Sources of Data

A. Primary Data

Primary data is collected directly by the researcher through surveys, interviews, observations, experiments, or sensors.

B. Secondary Data

Secondary data already exists and may come from institutional databases, government repositories, published datasets, research repositories, or public APIs.

C. Existing Public Datasets

Public datasets should be evaluated for source, license, quality, relevance, size, data collection context, and ethical restrictions. Public availability does not automatically make a dataset appropriate for every thesis.

VIII. Data Collection Considerations

1. Relevance

The data must directly relate to the research problem.

2. Sufficiency

There is no universal minimum dataset size. Students should consider whether they have sufficient and representative data for the proposed task.

3. Representativeness

The dataset should reasonably represent the population or situation being studied.

4. Accessibility

Students must determine whether the data can legally and practically be obtained, considering permissions, restrictions, and privacy requirements.

IX. Data Privacy and Ethical Considerations

All researchers must consider privacy, consent, security, and potential bias. Personally identifiable information should not be collected unless it is necessary and properly authorized.

Researchers should consider anonymization, de-identification, data minimization, and secure storage. A key principle is: Collect only the data necessary to achieve the purpose of the study.

PART II: DATA CLEANING

X. What is Data Cleaning?

Data cleaning is the process of identifying and correcting problems within a dataset. Real-world data commonly contains missing values, duplicate records, typographical errors, inconsistent formatting, incorrect data types, noise, outliers, and invalid values.

Poor-quality data can produce poor-quality models.

XI. Common Data Quality Problems

1. Missing Values

Missing values occur when information is absent. They may appear as blank cells, NULL, NaN, N/A, or Unknown.

Possible strategies include removing records, removing variables, or imputing values using approaches such as mean, median, or mode. The appropriate method depends on the nature and meaning of the data.

2. Duplicate Records

Duplicates may cause certain patterns to be overrepresented. Students should determine whether duplicates are accidental or legitimate repeated observations.

3. Inconsistent Data

The same category may be represented differently, such as Male, male, M, and MALE. Standardization may be necessary.

4. Incorrect Data Types

Data may be stored using inappropriate types. Students may need to convert strings to numerical values, dates, or appropriate categorical representations.

5. Outliers

Outliers are observations that differ substantially from other observations. They should not automatically be removed because unusual values may be meaningful rather than erroneous.

6. Noise

Noise refers to irrelevant or random information. In text, examples may include unnecessary symbols, repeated characters, HTML tags, or spam content.

XVII. Basic Data Cleaning Workflow

1. Inspect the dataset: number of records, variables, and data types.
2. Identify missing values and determine their extent.
3. Identify duplicate records.
4. Check data consistency and formatting.
5. Identify invalid or impossible values.
6. Apply appropriate cleaning procedures.
7. Document every important cleaning decision.

XVIII. Data Cleaning in Python – Refer to your laboratory week 3 copy

Pandas is commonly used for data processing in Python.

```
import pandas as pd
```

```
df = pd.read_csv("dataset.csv")
```

```
df.head()
```

```
df.info()
```

```
df.describe()
```

```
df.isnull().sum()
```

```
df.duplicated().sum()
```

```
df = df.drop_duplicates()
```

Students should understand the meaning of the output rather than simply executing the

commands. **PART III: EXPLORATORY DATA ANALYSIS**

XIX. What is Exploratory Data Analysis?

Exploratory Data Analysis (EDA) is the process of examining, summarizing, and visualizing a dataset to better understand its characteristics before major model development.

EDA helps answer questions such as: How is the data distributed? Are there missing values? Are classes balanced? Are some variables related? Are there unusual values?

XX. Why EDA is Important

EDA can reveal problems such as class imbalance. For example, a dataset with 95% of observations in one class may produce misleadingly high accuracy if a model simply predicts the majority class. EDA helps researchers identify such problems before interpreting model performance.

XXI. Descriptive Statistics

Mean

The mean is the average value.

Median

The median is the middle value after arranging observations.

Mode

The mode is the most frequently occurring value.

Minimum and Maximum

These values help identify the range of observations and may reveal unexpected values.

XXII. Data Visualization

1. Bar Chart

Useful for categorical data and class distributions.

2. Histogram

Useful for understanding the distribution of numerical variables.

3. Box Plot

Useful for identifying potential outliers and understanding data spread.

4. Scatter Plot

Useful for examining relationships between two numerical variables.

5. Correlation Matrix

Used to examine relationships among numerical variables. Correlation does not automatically imply causation.

XXIII. Class Distribution

For classification problems, students must examine how observations are distributed among classes. Class imbalance can affect model performance and make accuracy alone misleading.

PART IV: EDA FOR NLP DATA

XXIV. Exploratory Analysis of Textual Data

For NLP datasets, students should examine the number of documents, text length, empty responses, duplicate texts, frequently occurring words, language use, and class distribution.

Possible exploratory measures include character counts, word counts, sentence counts, and frequency analysis.

Text preprocessing may include lowercasing, removing unnecessary spaces, handling URLs, and removing duplicate texts. More advanced preprocessing such as tokenization, stop-word removal, stemming, and lemmatization should depend on the research objective and model.

XXV. The Iterative Relationship Between Collection, Cleaning, and EDA These processes are interconnected. A researcher may collect data, discover quality problems during cleaning, perform EDA, identify class imbalance, and then return to the data collection stage to obtain additional representative observations.

XXVI. Common Mistakes Students Should Avoid

1. Collecting data without a clear purpose.
2. Selecting an algorithm before understanding the problem and data.
3. Automatically deleting all missing data.
4. Treating all outliers as errors.
5. Ignoring class imbalance.
6. Cleaning data without documentation.

XXVII. Class Discussion Questions

1. What data does your proposed system actually need to perform its intended function?
2. Where will you obtain this data?
3. Do you have permission to access and use the data?
4. What will your model use as input?
5. What should your model predict, classify, generate, or retrieve?
6. What happens if a substantial portion of your dataset contains missing values?
7. How would you know whether your dataset is biased or unbalanced?

XXVIII. Week 3 Lecture Post-Requirement

Thesis Dataset Readiness Assessment

Each group will prepare a preliminary assessment of the data required for their proposed thesis.

- A. Thesis Title
- B. Proposed AI/Computational Problem
- C. Required Dataset

- D. Data Source
- E. Initial Features or Input Variables
- F. Target Variable or Expected Output
- G. Potential Data Issues
- H. Data Ethics and Privacy Statement

XXIX. Expected Week 3 Deliverables

Deliverable 1: Thesis Dataset v1.0

A preliminary version of the dataset relevant to the student's thesis.

Deliverable 2: Data Dictionary

Variable	Description	Type	Role
GPA	Student grade average	Numerical	Feature
Attendance	Attendance percentage	Numerical	Feature
Risk	Academic risk status	Categorical	Target

Deliverable 3: Data Cleaning Report

- Number of original records
- Missing values identified
- Duplicate records identified
- Cleaning procedures performed
- Final dataset size

Deliverable 4: Exploratory Data Analysis Report

- Basic descriptive statistics
- At least three relevant visualizations
- Key observations and findings

XXX. Key Takeaways

1. Start with the research problem rather than immediately selecting an algorithm.
2. Define the data requirements needed to address the problem.
3. Data quality matters because incorrect, biased, missing, or poorly labeled data can affect model performance.
4. Explore the data before modeling to understand its characteristics and limitations.
5. Document the complete process for research reproducibility and thesis writing.

"A Machine Learning model does not begin with an algorithm. It begins with a problem and the data needed to understand that problem. Before asking which model performs best, a researcher must first understand what the data represents, where it came from, and whether it is suitable for the task."

The outputs from Week 3 will serve as the foundation for Week 4: Introduction to Machine Learning and the Basic ML Workflow. Students will use their preliminary datasets to begin identifying appropriate Machine Learning tasks and developing baseline models.